Shield Band: A Sustainable Forensic-Grade Personal Safety Wearable

Submission for AURA 2K26 | Theme: Sustainability

1. Methodology

The Shield Band is an embedded wearable safety system centered on an ESP32-CAM microcontroller, a GPS module, a microphone, an OLED display, and a tactile SOS push-button. The design follows a privacy-first operating principle: the camera and microphone remain inactive during normal operation and are activated only after a deliberate SOS trigger. To align with the competition's Sustainability theme, the wearable housing is proposed to use biodegradable mycelium composites or flax-reinforced PLA, reducing dependence on conventional plastic and supporting a lower-impact end-of-life pathway. The electronics operate in a low-power standby state to reduce energy consumption, while a small solar harvester can provide supplementary charging. The ESP32-CAM is selected for its compact form factor and cost-to-performance balance, supporting an affordable prototype suitable for students, commuters, and other users who need rapid access to an emergency trigger. The OLED provides immediate local feedback such as system readiness, alert status, and recording state.

2. Proposed Implementation

When the user presses the physical SOS button, the ESP32-CAM exits its low-power state and initiates the emergency workflow. The GPS module acquires the current geographical coordinates, while the controller activates the camera and microphone to capture time-linked audio-visual evidence. A GSM/LTE communication module is included in the proposed architecture to provide cellular connectivity for an automated SMS alert containing a Google Maps location link to pre-configured emergency contacts. Separately, where Internet connectivity is available, the ESP32-CAM transmits the emergency metadata and captured evidence to a secure cloud platform using an encrypted connection. Authorized campus safety or response personnel can access the resulting dashboard for location information and cloud-stored evidence. The system therefore separates the immediate cellular alert path from the Internet-based evidence and dashboard path, improving clarity of the communications architecture.

3. Proposed Outcome of the Project

The primary outcome is a low-cost, scalable wearable safety device that bridges the gap between an emergency trigger and subsequent response or investigation. The proposed target is a prototype cost below approximately ₹2,500, subject to component and manufacturing prices, to explore accessibility for students, women, children, night-shift workers, and other users who may require a rapidly accessible emergency mechanism. The project also supports the broader objective of Sustainable Development Goal 11 (Sustainable Cities and Communities) by combining personal safety infrastructure with resource-conscious hardware design.

- **Smartphone-Independent Safety:** Users can initiate an emergency workflow without first unlocking a smartphone or depending on mobile-app permissions.

• **Privacy-First Evidence:** Audio-visual capture is event-triggered rather than continuous, reducing unnecessary routine surveillance while creating time-linked material for post-incident review.

• **Location-Aware Response:** GPS coordinates can be transmitted with the emergency alert so that responders receive a specific geographical location.

• **Sustainable Design:** Biodegradable or bio-reinforced housing, low-power operation, and supplementary solar harvesting are incorporated into the proposed design.

• **Future Roadmap — Shield AI:** A future AI layer could identify nearby verified safe zones such as police stations and hospitals and provide route guidance in real time.

4. Proposed Block Diagram

The architecture below separates the emergency alert path, cloud evidence path, local user feedback, sustainable power subsystem, and privacy control. A GSM/LTE module is shown explicitly because an ESP32-CAM alone does not provide cellular SMS connectivity.

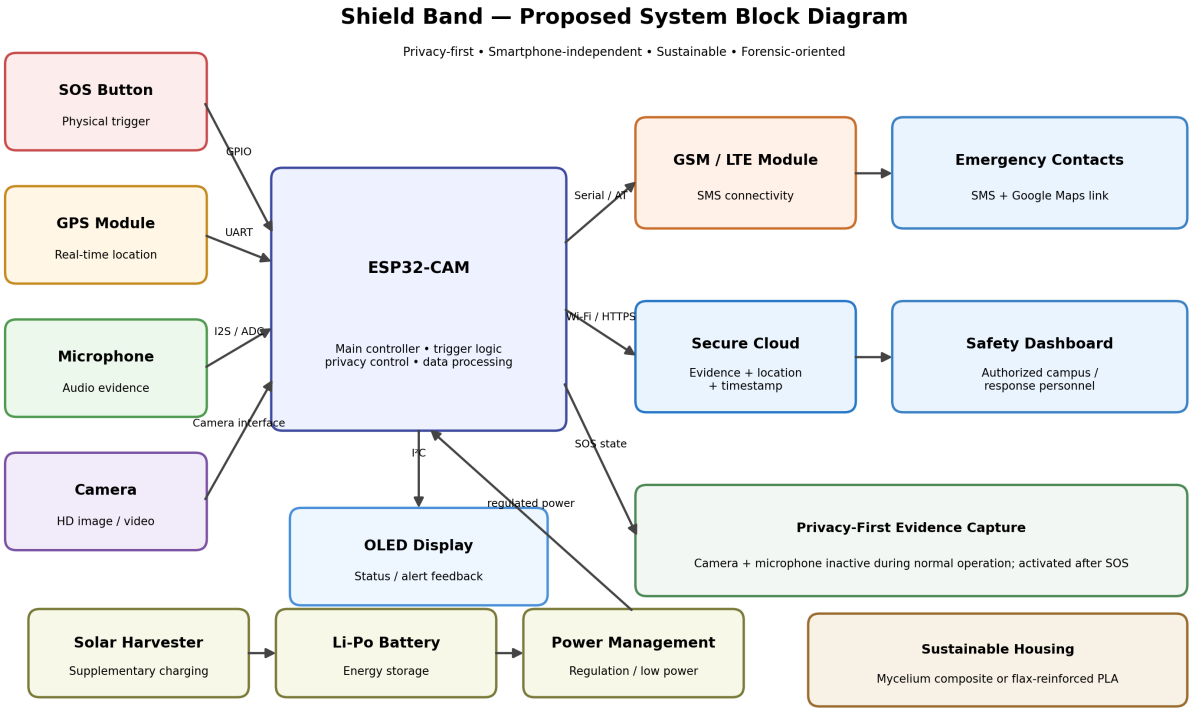


Figure 4.1. Proposed Shield Band system architecture.

Technical and Legal Note

The system is intended to support preservation and handling of electronic information for post-incident review. Recording audio-visual material does not, by itself, establish legal admissibility; evidentiary acceptance depends on applicable Indian law, authenticity, integrity, provenance, and the circumstances of collection and presentation. Accordingly, the prototype should emphasize secure timestamps, controlled access, and integrity-preserving storage rather than claiming automatic legal admissibility.